

RL4HCI: Reinforcement Learning for Humans, Computers, and Interaction

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ABSTRACT

Reinforcement learning (RL) is emerging as an approach to understand intelligence in both humans and machines. However, if RL is to have a meaningful impact in human–computer interaction, it is critical that these two threads are integrated. This is required for genuinely interactive RL-based systems which take into account user capacities and preferences. This workshop will build a community and form a research agenda for investigating RL in HCI.

CCS CONCEPTS

• **Human-centered computing** → **HCI theory, concepts and models**; **User models**.

KEYWORDS

reinforcement learning, interactive Artificial Intelligence, model-based/model free, POMDP, MDP, cognitive models, applications

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1 BACKGROUND

In computer science, reinforcement learning (RL) is the problem faced by an agent that must learn how to act in a sequential decision environment. Recent examples where RL has been used for *modelling users* include multitasking [9, 13] visual search [5], decision making [1, 6], and gestures [4]. RL has also been investigated to understand interaction from sensor data, such as for inferring everyday routines [2] or parameters of cognitive models [16]. These

models assume that users' behavior can be approximated as a computationally rational policy given the assumed preferences and capacity limits of the user [10, 19]. In other words, an RL algorithm can be used to find human-like interaction strategies given the preferences and capacity limits of the user. Viewed as an RL problem, hand-eye coordination, for example, is a consequence of optimal adaptation to manual, oculomotor and visual perception constraints.

RL is also attracting interest as a principle of adaptive interfaces and computational design. Examples of RL *intelligent interaction problems* include dialogue management [25], information retrieval [12], intelligent tutoring [23], adaptive interface problems [8], and the interface optimization problem [14, 15, 17, 18, 24]. Related research includes work on the movement interaction design problem [11], work on machine learning for the detection, diagnosis and treatment of mental health [27], work on the problem of incorporating human advice into reinforcement learning [21], and on human-AI team performance [3], and work on the human-robot interaction (HRI) problem [7, 20, 22]. More generally work on intelligent interaction problems includes work on explanation in AI [26].

The main objective of this online workshop is to engage serious discussion at the intersection of these two threads. If RL is to have a meaningful impact on HCI then it is critical to develop a unified understanding of the intersection of human and machine viewpoints. Doing so will allow RL for HCI to develop its own research agenda for exploiting machine learning to address outstanding issues in human-centred computing. Such an agenda might then support design and genuinely adaptive personalisation to individual preferences and capabilities. It might also offer a unifying framework for describing human–computer and human–AI interaction as optimal sequential decision making.

With this ambition in mind, the online workshop will aim to deliver (1) a survey which defines terms and reviews relevant work and (2) a research agenda for developing RL for human–computer interaction. It will also aim to extend an existing and growing community of scholars who work in the intersection between HCI and AI. Topics for discussion will be largely driven by attendees but will likely include: model inference, optimal experimental design, and how to define problems with human-like bounds. In addition, we will discuss evaluation: While there are a number of methods for running simulations to evaluate RL algorithms, these simulations

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may not reflect how well an RL-based system will perform once it is in the world and interacting with real people.

The online workshop will build on previous workshops at CHI¹, a series of summer schools on “Computational Interaction” funded by ACM² and a Dagstuhl meeting.³

2 ORGANIZERS

Contact: Andrew Howes (andrew.howes@aalto.fi).

Dorota Glowacka Dorota Glowacka is an assistant professor in Artificial Intelligence and Machine Learning in the Department of Computer Science, University of Helsinki, Finland, where she leads the Exploratory Search and Personalization (ESP) group. From June 2017 to December 2018, she was an assistant professor in the Institute for Language, Cognition and Computation in the School of Informatics, University of Edinburgh, and from 2011 to 2017 she was a researcher at the Helsinki Institute for Information Technology (HIIT), Finland. She completed a Ph.D. in Machine Learning at University College London (UCL). Her research focuses on exploratory search, interactive information retrieval and reinforcement learning. She regularly publishes at CIKM, CHIIR and IUI, where she received best paper award in 2013. In recent years, she gave three tutorials on the application of bandit algorithms and reinforcement learning in information retrieval and recommender systems at RecSys.

Andrew Howes is Visiting Professor of Cognitive Science at Aalto University and Professor of HCI at the University of Birmingham. He has previously been an Associate Editor for Human-Computer Interaction, an Associate Editor for Cognitive Science journal and the Marshall C. Weinberg Visiting Professor of Cognitive Science at University of Michigan. His research interests are in computer models of human decision making.

Jussi Jokinen is a research fellow at University of Helsinki and Finnish Center for Artificial Intelligence. He received his PhD in cognitive science in 2015 from University of Jyväskylä, Finland. His work deals with computational cognitive models, reinforcement learning, and emotion.

Antti Oulasvirta leads the User Interfaces research group at Aalto University and the Interactive AI research program at FCAI (Finnish Center for AI). Prior to joining Aalto, he was a Senior Researcher at the Max Planck Institute for Informatics and the Cluster of Excellence on Multimodal Computing and Interaction at Saarland university. He received his doctorate in Cognitive Science from the University of Helsinki in 2006, after which he was a Fulbright Scholar at the School of Information in University of California-Berkeley in 2007-2008 and a Senior Researcher at Helsinki Institute for Information Technology HIIT in 2008-2011. He is a member of ELLIS (European Laboratory for Learning and Intelligent Systems).

Özgür Şimşek is a senior lecturer (associate professor) at the University of Bath, in the Department of Computer Science, where she leads the Artificial Intelligence Research Group. Earlier, she was a research scientist at the Center for Adaptive Behaviour and Cognition at the Max Planck Institute for Human Development. She received her PhD in Computer Science in 2008 from the University of Massachusetts Amherst. Her research spans a broad range

of areas in machine learning, including reinforcement learning, supervised learning, and learning from small data sets. Her applied and theoretical work in machine learning has appeared in leading venues, including Proceedings of the National Academy of Sciences, KDD, ICML, and NeurIPS. She is Co-Investigator of UKRI Doctoral Training Centre for Accountable, Responsible, and Transparent Artificial Intelligence (ART-AI), where she is a member of the Management Team and Theme Leader in Machine Learning.

3 LINK TO WEBSITE

<https://sites.google.com/view/rl4hci/home>

4 PRE-WORKSHOP PLANS

Our goal is to engage participants prior to the online workshop to form views on key research problems and use the workshop to share views and form a joint agenda.

- Advertise the online workshop and contact potential attendees and sponsors. We will reach out to relevant communities outside of HCI. We will also encourage submissions from PhD students, and work that’s not yet “complete”.
- Request submission of 2-3 page position papers.
- Select participants. Emphasis is on participants who already know the basics of RL. This is a requirement based on our previous experience; The workshop will not be a tutorial.
- Establish a diverse committee of 10 to 20 key people and ask them to brainstorm the research questions that will be addressed during the workshop.
- Before the workshop we will pick pairs or groups based on themes identified in the position papers and ask the groups to meet before the workshop and record video statement to the rest of the attendees.
- Prepare the online venue and distribute connection information and workshop structure to attendees.
- We will request prerecorded talks (with the exception of keynotes). Talks will be prerecorded and shared one week before the online workshop, and we will request that participants watch the videos. We will divide talks into shorter (lightning talks 2-5 mins) and a bit longer talks (10 mins).

5 WORKSHOP STRUCTURE

The workshop will consist of three blocks:

- Block 1: A mix of long (1-2 invited keynotes), medium and short (lightning) online talks. Morning session focuses on keynotes and Q&A for prerecorded videos.
- Block 2: Mid-session: Focuses on the themed pre-workshop activities. Groups lead online discussion on those themes. (This is the “bottom up” part of workshop). Online multimedia breakout groups will be organized that use a combination of audio communication and high-bandwidth online document creation.
- Block 3: Afternoon session: An online co-writing session that focuses on forming a research agenda or structure for a review paper. This activity will use an online writing tool.

¹<https://hccompmodeling.wordpress.com/>

²<https://www-edc.eng.cam.ac.uk/summerschool/>

³<https://userinterfaces.aalto.fi/dagstuhl-seminar-on-computational-interaction/>

6 POST-WORKSHOP PLANS

- Write an RL4HCI manifesto (research vision and agenda) on the basis of the content created during the workshop.
- The manifesto will be submitted to one of the following venues: Interactions, ACM Communications, AAAI Magazine, Interactions, HCI journal, Nat Mac Int etc.
- Encourage a small group to organize a follow-on workshop proposal for ACM CHI 2022.
- Maintain an RL4HCI email list that provides quarterly updates on relevant events.

7 250-WORD CALL FOR PARTICIPATION

Call for participation in an online workshop on RL4HCI. In computer science, reinforcement learning (RL) is the problem faced by an agent that must learn a sequential environment through trial and error. Recently, reinforcement learning algorithms have emerged as a key technology for a range of purposes including modeling people and machine intelligence. However, if reinforcement learning is to have a meaningful impact in human-computer interaction then it is critical that these two threads are integrated. This is required for genuinely interactive RL-based systems which take into account user capacities and preferences. Interactive RL systems will need to model a user's visual search, multitasking, decision making, emotions, social connectivity etc. They will also need to use these models to assist with personalisation, tutoring, medical diagnosis, and interaction design to name but 4 possible application domains.

The goals of this 1-day workshop will be to build a community and form a research agenda for investigating reinforcement learning problems in human-computer interaction. Position papers of 2-3 pages are requested in ACM format. PhD students, along with all levels of experience, are encouraged to apply. A key requirement for acceptance will be evidence of knowledge of RL and interest in HCI. Submissions should be sent to Andrew Howes (Andrew.Howes@Aalto.fi) before noon Pacific Standard Time, Friday February 19th 2021. At least one author of each accepted position paper must attend the online workshop and all participants must register for both the workshop and for at least one day of the conference. Note that a special remote attendance rate will be announced.

Website: <https://sites.google.com/view/rl4hci/home>

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